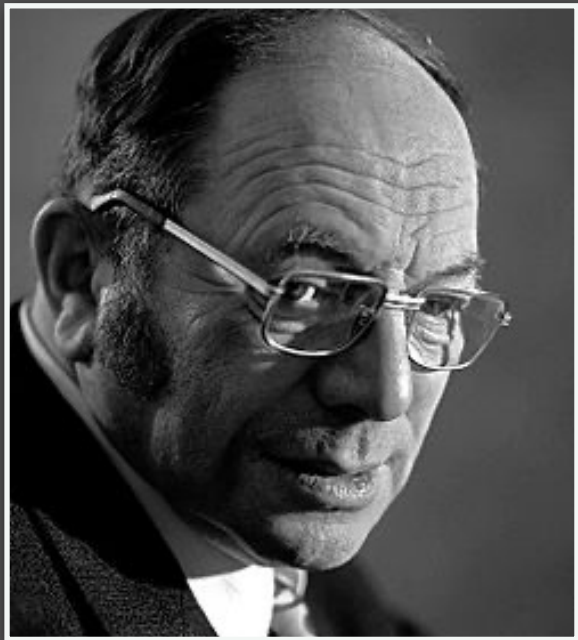


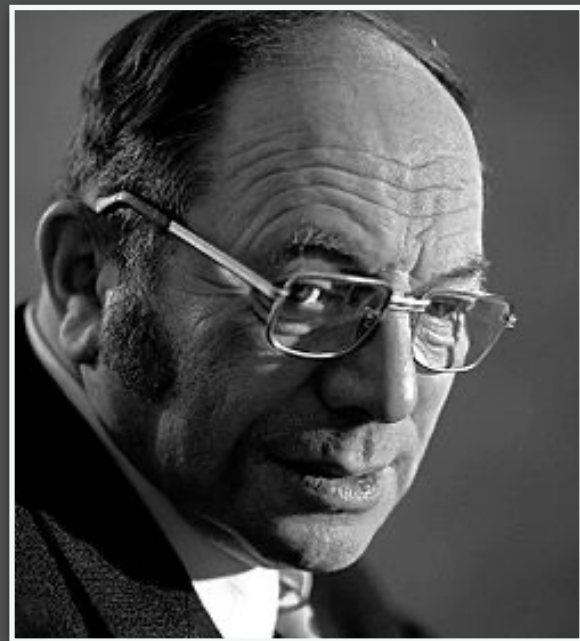


Kantorovich's Problem:

(Discrete version)

$$\min_{\pi \in \Pi(\mathbf{a}, \mathbf{b})} \sum_{i=1}^n \sum_{j=1}^m C_{i,j} \pi_{i,j}$$

Kantorovich is an LP



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Matrix/vector form:

$$\begin{aligned} \min_{\pi \in \mathbb{R}^{n \times m}} \quad & \langle C, \pi \rangle = \text{tr}(C^\top \pi) \\ \text{s.t.} \quad & \pi \mathbf{1}_m = a, \\ & \pi^\top \mathbf{1}_n = b, \\ & \pi \geq 0. \end{aligned}$$

$$\min_{\pi \in \mathbb{R}^{n \times m}} \sum_{i=1}^n \sum_{j=1}^m C_{ij} \pi_{ij}$$

Linear objective

$$\text{s.t.} \quad \sum_{j=1}^m \pi_{ij} = a_i \quad \forall i = 1, \dots, n,$$

$$\sum_{i=1}^n \pi_{ij} = b_j \quad \forall j = 1, \dots, m,$$

$$\pi_{ij} \geq 0 \quad \forall i, j.$$

Linear equality constraints

Vectorized form [$x = \text{vec}(\pi)$]:

$$\begin{aligned} \min_{x \in \mathbb{R}^{nm}} \quad & c^\top x \\ \text{s.t.} \quad & (I_n \otimes \mathbf{1}_m^\top) x = a, \\ & (\mathbf{1}_n^\top \otimes I_m) x = b, \\ & x \geq 0. \end{aligned}$$

Q: How many variables and constraints?

Particular Case: Uniform Measure